

Ex. No:6 Use to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

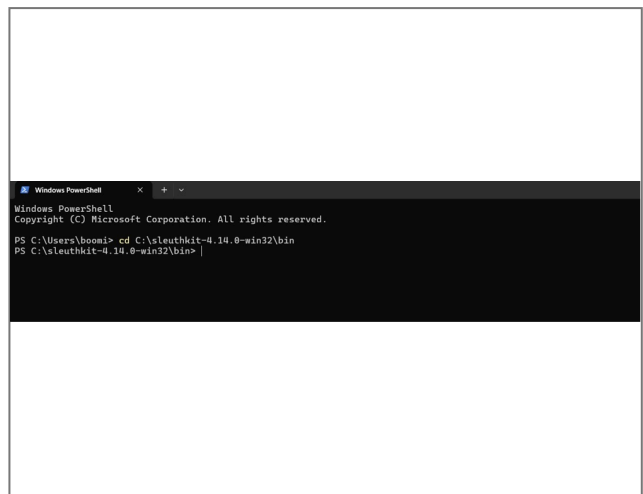
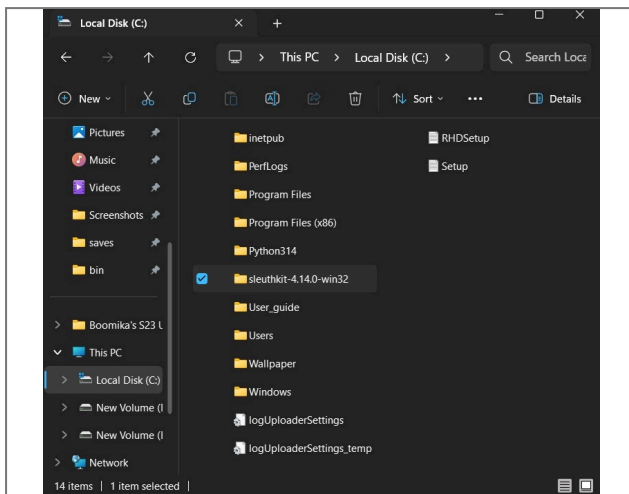
Download Sleuth Kit:

Visit the official Sleuth Kit website or

https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v and download the latest version for Windows.

Install Sleuth Kit:

Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.



Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

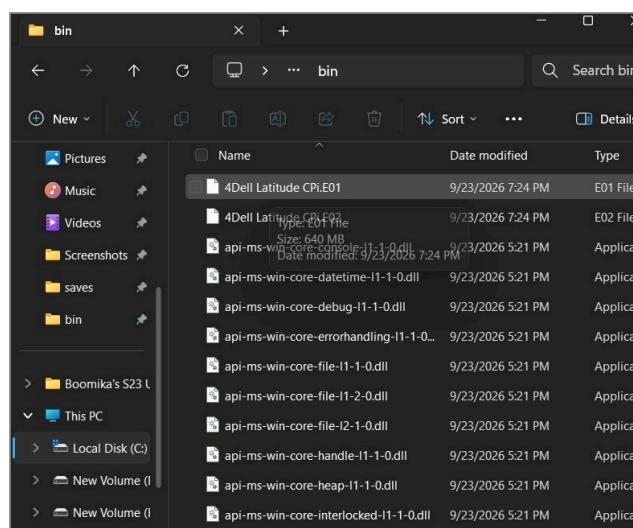
Create Disk Image:

Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.

Download the below files from the google drive

4Dell Latitude Cpi.E01

4Dell Latitude Cpi.E02



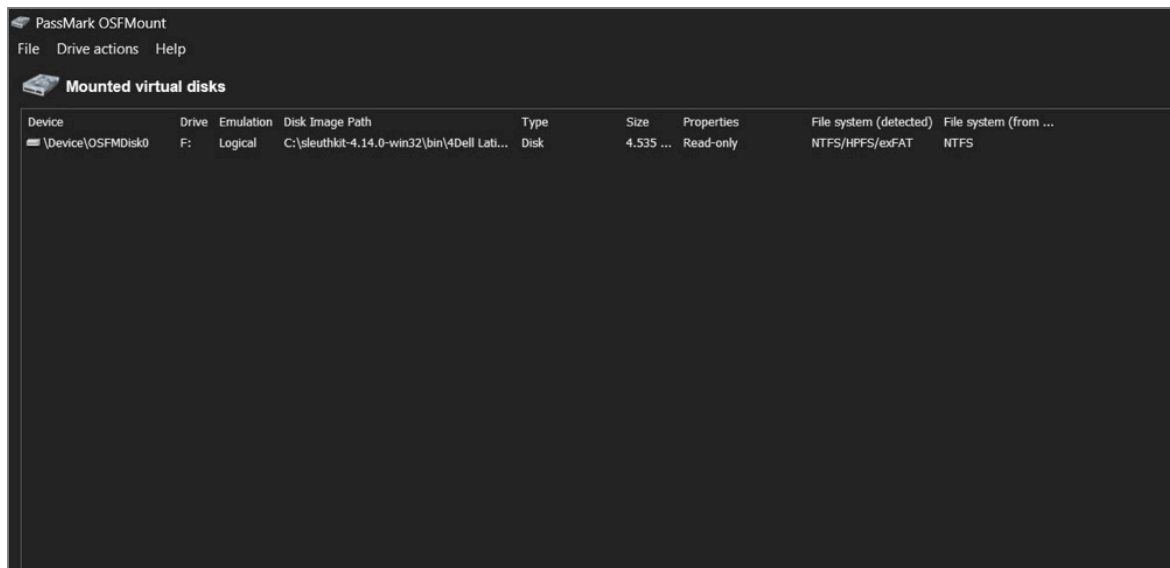
Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

Mount the Image:

You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.

This step is optional but helps with navigating the file system easily.



Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

Navigate to the Sleuth Kit Directory:

Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

Identify File System Type with fsstat:

Run the command:

```
fsstat [image file] > filesystem_info.txt
```

This command outputs information about the file system, which is crucial for understanding the structure of the disk.

List Partitions with mmls:

Run the command:

```
mmls [image file] > partitions.txt
```

This command lists the partitions within the image file.

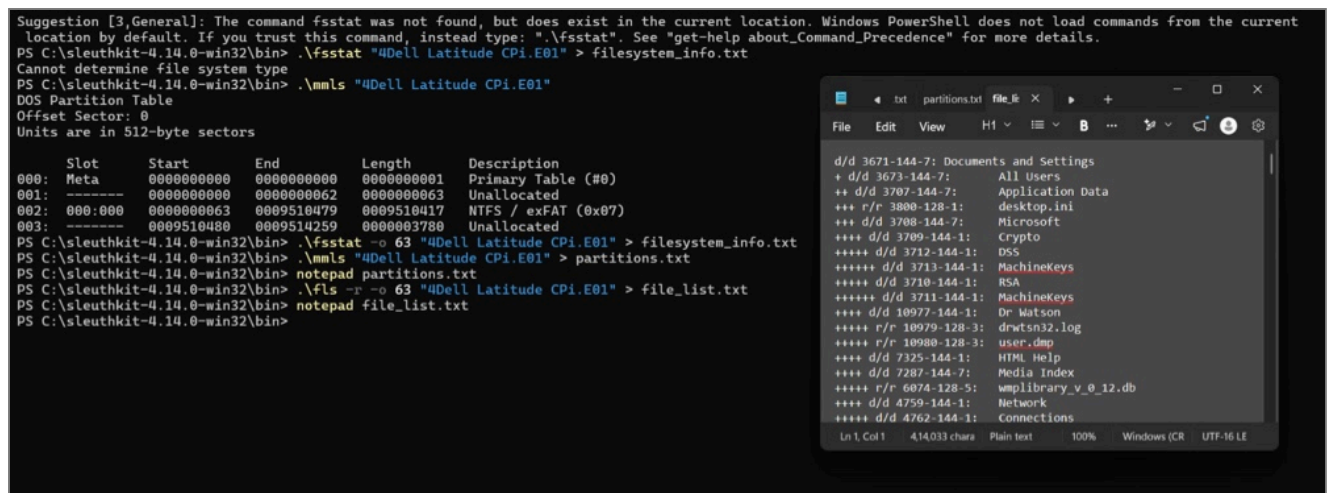
```
Slot      Start      End          Length      Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  -----  0000000000  0000000062  0000000063  Unallocated
002:  000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:  -----  0009510480  0009514259  0000003780  Unallocated
PS C:\sleuthkit-4.14.0-win32\bin> .\fsstat -o 63 "4Dell Latitude CPi.E01" > filesystem_info.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\mmls "4Dell Latitude CPi.E01" > partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> |
```

Analyze File System with fls:

Run the command:

```
fls -r [image file] > file_list.txt
```

This command recursively lists files and directories in the file system, showing their metadata.



```
Suggestion [3,General]: The command fsstat was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\fsstat". See "get-help about_Command_Precedence" for more details.
PS C:\sleuthkit-4.14.0-win32\bin> .\fsstat "4Dell Latitude Cpi.E01" > filesystem_info.txt
Cannot determine file system type
PS C:\sleuthkit-4.14.0-win32\bin> .\mmls "4Dell Latitude Cpi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length    Description
000: Meta      0000000000 0000000000 0000000001 Primary Table (#0)
001: ----- 0000000000 0000000062 0000000063 Unallocated
002: 000:000 0000000063 0009510479 0009510417 NTFS / exFAT (0x07)
003: ----- 0009510480 0009514259 0000003780 Unallocated

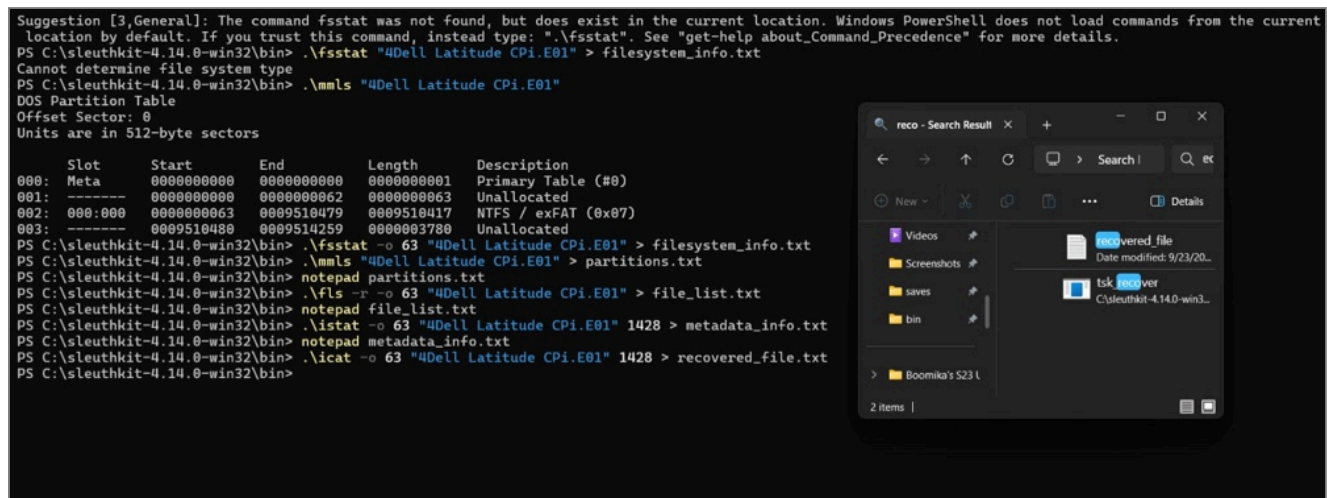
PS C:\sleuthkit-4.14.0-win32\bin> .\fsstat -o 63 "4Dell Latitude Cpi.E01" > filesystem_info.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\mmls "4Dell Latitude Cpi.E01" > partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\fls -r -o 63 "4Dell Latitude Cpi.E01" > file_list.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad file_list.txt
PS C:\sleuthkit-4.14.0-win32\bin>
```

Recover Deleted Files with icat:

To extract a specific file, use icat:

```
icat [image file] [inode number] > [output file]
```

Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.



```
Suggestion [3,General]: The command fsstat was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\fsstat". See "get-help about_Command_Precedence" for more details.
PS C:\sleuthkit-4.14.0-win32\bin> .\fsstat "4Dell Latitude Cpi.E01" > filesystem_info.txt
Cannot determine file system type
PS C:\sleuthkit-4.14.0-win32\bin> .\mmls "4Dell Latitude Cpi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length    Description
000: Meta      0000000000 0000000000 0000000001 Primary Table (#0)
001: ----- 0000000000 0000000062 0000000063 Unallocated
002: 000:000 0000000063 0009510479 0009510417 NTFS / exFAT (0x07)
003: ----- 0009510480 0009514259 0000003780 Unallocated

PS C:\sleuthkit-4.14.0-win32\bin> .\fsstat -o 63 "4Dell Latitude Cpi.E01" > filesystem_info.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\mmls "4Dell Latitude Cpi.E01" > partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad partitions.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\fls -r -o 63 "4Dell Latitude Cpi.E01" > file_list.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad file_list.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\istat -o 63 "4Dell Latitude Cpi.E01" 1428 > metadata_info.txt
PS C:\sleuthkit-4.14.0-win32\bin> notepad metadata_info.txt
PS C:\sleuthkit-4.14.0-win32\bin> .\icat -o 63 "4Dell Latitude Cpi.E01" 1428 > recovered_file.txt
PS C:\sleuthkit-4.14.0-win32\bin>
```

Step 5: Analyze Metadata

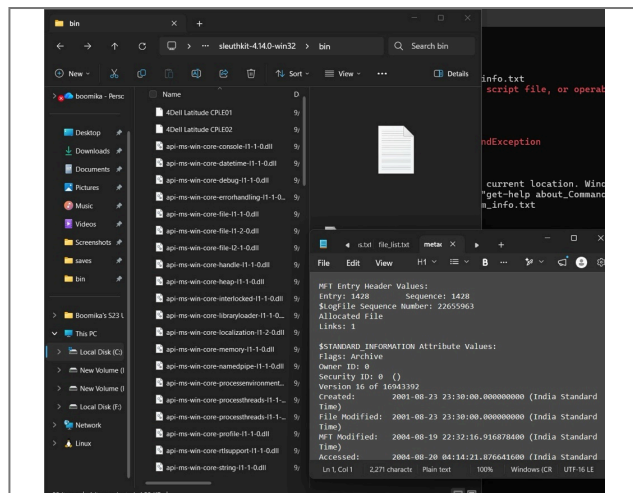
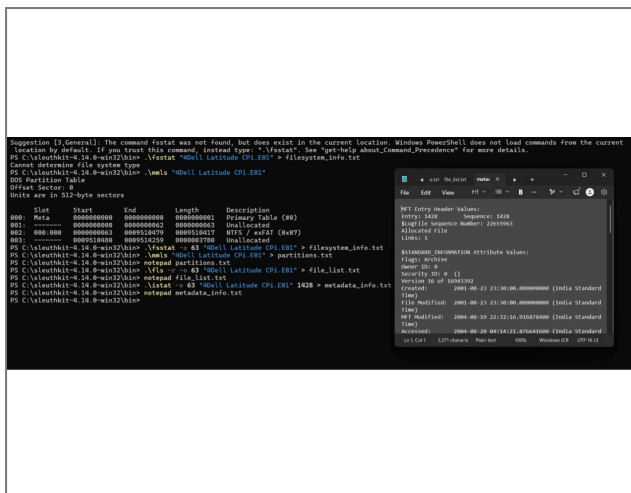
Extract metadata from files to understand more about the file's history.

View Metadata with istat:

Run the command:

```
istat [image file] [inode number] > metadata_info.txt
```

This provides detailed information about a file, including timestamps, size, and allocation status.



Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

Create Timeline with mactime:

Generate a body file using fls:

```
fls -m / -r [image file] > body.txt
```

Then create the timeline:

```
mactime -b body.txt > timeline.txt
```

The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

Compile the Data:

Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

Analyze and Document:

Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

Archive Evidence:

Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

Store Securely:

Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.